

Khaled Al Tamimi

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WORK EXPERIENCE

Software Engineering Intern | German Service Center – Dammam, Saudi Arabia *July 2025 – August 2025*

- Designed and implemented an automated invoice estimation workflow, calculating service estimates, routing them for managerial review and approval, and automatically sending approved estimates to customers via email.
- Developed a part-sourcing algorithm to estimate procurement timelines and identify optimal vendors when parts were out of stock.
- Wrote Python automation scripts to perform daily database checkups, clean outdated records, and generate operational metrics such as average repair time and recurring issues.
- Drafted comprehensive technical documentation and system architecture workflows for non-technical stakeholders, ensuring smooth long-term software maintenance.

PROJECTS

Network Anomaly Detection & Threat Detection Engine | Python, scikit-learn, FastAPI, Docker, AWS *In progress*

- Architecting an end-to-end ML pipeline (data → model → deployed API) on the NSL-KDD intrusion-detection dataset to bridge a security background with applied AI/ML.
- Built the ingestion, profiling, and feature-engineering pipeline, including a persisted categorical encoder for reusable inference.
- Sequenced model development deliberately: binary before multiclass classification, precision/recall/F1 over raw accuracy given class imbalance, and a RandomForest baseline before XGBoost/LightGBM.
- Roadmapping a Dockerized FastAPI scoring endpoint for deployment on AWS EC2, documenting each design tradeoff alongside the code

Grade Vault | JavaScript, React, Node.js, Express, SQLite *March 2026*

- Built a role-based grade management system with strict identity and access management (student/teacher/admin roles) using server-side middleware.
- Leveraged and audited open-source middleware to implement 9 CWE-mapped security controls including session management, password strength enforcement and CSV validation.
- Designed secure authentication using HTTP-only session cookies, ensuring safe handling of sensitive user credentials and academic records.

EDUCATION

Bachelor of Computer Science | University of New Brunswick, Fredericton, NB *Sept 2023 – Present*

- Dean's List 2023-2024 | GPA: 3.5
- Relevant Coursework: Net-Centric Computing, Software Security, Foundations of Privacy, Cryptanalysis & Database Security, Computer Architecture, Software Engineering, Algorithm Design and Analysis, Operating Systems

TECHNICAL SKILLS

- **Languages & Databases:** JavaScript, Java, Python, C, MySQL, SQLite
- **Frameworks & Libraries:** React, Node.js, Express, JavaFX, Apache POI
- **Security:** Incident Triage & Escalation, IAM, Session Management, Threat Modeling, Cryptographic Integrity (AES-256, SHA-256), Vulnerability Scanning.
- **Tools & Testing:** Docker, Git/GitHub, Wireshark, Jupyter, JUnit, REST APIs, Microsoft Excel (advanced formatting, pivot tables)
- **Cloud:** AWS EC2, Cloudflare